

Enterprise Cyber Threat Intelligence & SOC Monitoring Dashboard

1. Executive Summary

This project focuses on building an interactive enterprise-style cybersecurity dashboard using Power BI to monitor cyber threats, analyse suspicious network activity, and visualize global attack patterns.

The dashboard simulates a real-world Security Operations Center (SOC) environment where security teams can track attacks, identify high-risk threats, monitor malicious traffic behaviour, and gain actionable insights through interactive visualizations.

The project combines cybersecurity analytics, business intelligence, and modern dashboard storytelling to create a professional SOC monitoring solution.

2. Problem Statement

Background

Organizations generate huge amounts of network traffic every day. Monitoring this traffic manually becomes extremely difficult because cyber-attacks are increasing rapidly and attackers use different methods to hide malicious activities.

Traditional reports are often static and difficult to analyse quickly. Security teams need a centralized and interactive platform to monitor threats efficiently.

Objective

The objective of this project is to develop a modern cybersecurity intelligence dashboard that can:

- Monitor cyber attack trends
- Analyse malicious network traffic
- Identify critical and high-risk threats
- Visualize global cyber-attacks
- Support threat investigation using interactive dashboards
- Simulate a real-world SOC monitoring environment

Scope of the Project

The dashboard focuses on:

- Network traffic analysis
 - Attack category monitoring
 - Protocol and service analysis
 - Threat severity tracking
 - Geographic attack visualization
 - Risk monitoring and investigation
 - Interactive cybersecurity reporting using Power BI
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3. Dataset Used

Primary Dataset

The project uses the **UNSW-NB15 Cybersecurity Dataset** sourced from Kaggle.

The dataset contains:

- Network traffic sessions
- Attack categories
- Protocol information
- Services used during attacks
- Malicious traffic behaviour
- Network activity metrics

Dataset Link

[UNSW-NB15 Kaggle Dataset](#)

4. Additional Data Simulation & Enhancements

To make the dashboard more realistic and enterprise-ready, additional simulated fields and enhancements were created during data preparation.

The following improvements were added:

- Generated realistic Source IP Addresses
- Generated realistic Destination IP Addresses
- Created global Region and Country mappings
- Simulated attack timings based on attack behaviour
- Created Risk Level classifications (Low, Medium, High, Critical)
- Added attack investigation logic for SOC monitoring
- Designed realistic cyber threat storytelling visuals

These enhancements helped transform the dataset into a realistic enterprise cybersecurity monitoring environment.

5. Tools & Technologies Used

Tool	Purpose
Microsoft Power BI	Dashboard Development & Visualization
Power Query	Data Cleaning & Transformation
DAX	KPI Calculations & Analytics
Excel / CSV Files	Dataset Source

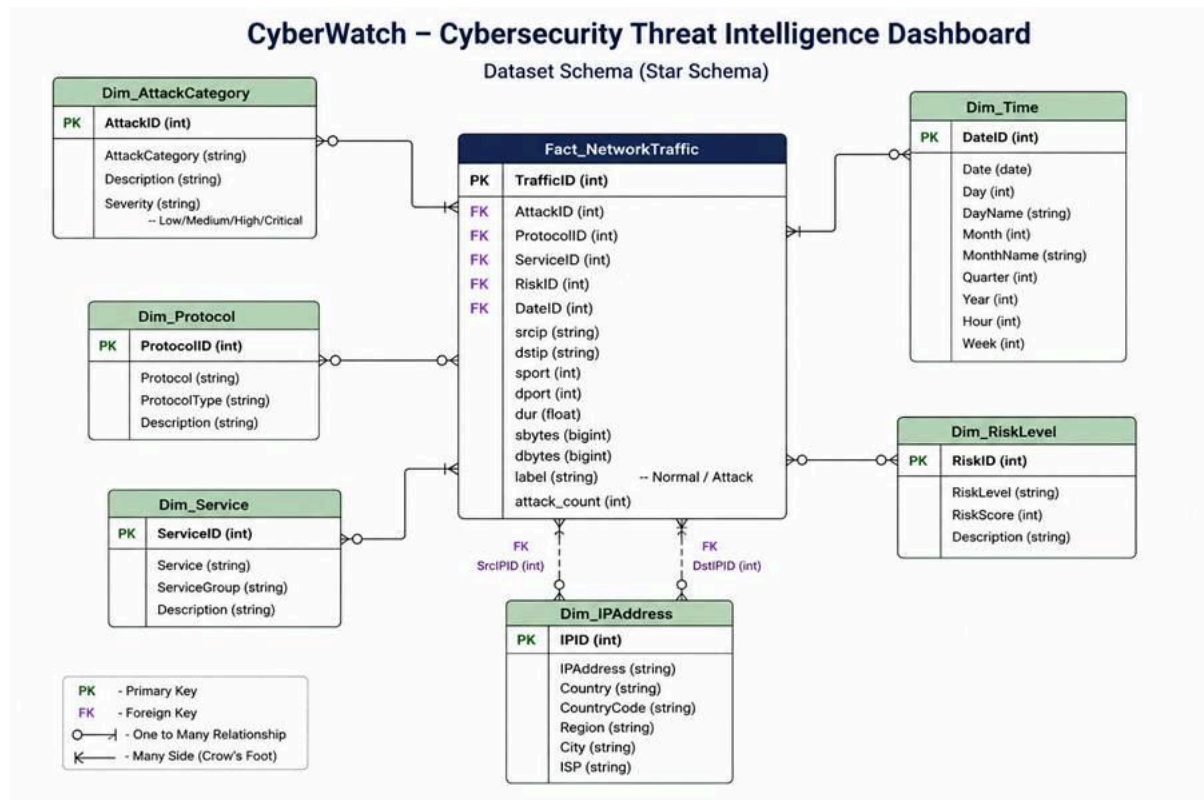
6. Data Preparation & Modelling

Data Cleaning & Transformation

The following preprocessing steps were performed:

- Cleaned missing and inconsistent values
- Removed duplicates
- Standardized attack categories
- Converted data types
- Created threat severity classifications
- Created region mapping tables
- Prepared data for visualization using Power Query

Data Modelling



A Star Schema model was created using:

Fact Table

- Fact_NetworkTraffic

Dimension Tables

- Dim_AttackCategory
- Dim_Protocol
- Dim_RiskLevel
- Dim_Region
- Dim_Date

The model supports dynamic filtering and interactive analysis across the dashboard.

7. Dashboard Pages Developed

Page 1 — Home / Command Center

Purpose:

- Dashboard landing page
- Navigation center for all pages
- Executive introduction to the project

Features:

- Interactive navigation buttons
- Cybersecurity themed UI
- SOC-style dashboard design

Page 2 — Threat Overview Dashboard

Purpose:

Analyse overall cyber attack activity and network behaviour.

Visuals Included:

- KPI Cards

- Attack Trend Analysis
 - Attack Distribution by Protocol
 - Threat Severity Distribution
 - Attack Category Analysis
 - Top Targeted Services
 - Executive Threat Insights
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Page 3 — Geo Intelligence Dashboard

Purpose:

Visualize global attack activity and regional threat patterns.

Visuals Included:

- Global Threat Heat Map
 - Top Countries by Attacks
 - Attacks by Region
 - Regional Attack Trends
 - Top Targeted Services by Region
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Page 4 — Risk Monitoring & Investigation

Purpose:

Monitor critical threats and support cyber incident investigation.

Visuals Included:

- Critical Threat KPIs
- High Risk Threat Monitoring
- Incident Severity Analysis
- Attack Behaviour by Time
- Threat Investigation Tables

8. KPI Measures Created

The dashboard includes several interactive KPI measures such as:

- Total Network Events
 - Total Attacks
 - Attack Rate %
 - Critical Threat Count
 - High Risk Threat Count
 - Unique IP Addresses
 - Most Targeted Service
 - Most Targeted Protocol
 - Average Attacks Per Day
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9. Key Features of the Dashboard

- Enterprise SOC-style UI Design
 - Multi-page Interactive Dashboard
 - Cyber Threat Heatmap Visualization
 - Dynamic KPI Monitoring
 - Risk-Based Threat Analysis
 - Interactive Filtering & Slicers
 - Executive-Level Cybersecurity Storytelling
 - Dark Theme Cybersecurity Interface
 - Simulated Real-World SOC Monitoring Experience
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10. Expected Outcomes

The final dashboard helps organizations:

- Improve cyber threat visibility
- Identify high-risk attack patterns quickly
- Monitor suspicious network behaviour
- Support security investigation teams
- Improve cybersecurity decision-making
- Analyse global attack activity interactively

11. Challenges Faced

Some challenges during development included:

- Cleaning large cybersecurity datasets
 - Creating realistic threat simulation logic
 - Designing enterprise-level dashboard layouts
 - Maintaining clean UI without overcrowding visuals
 - Simulating realistic SOC monitoring behaviour
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12. Conclusion

This project demonstrates how Power BI can be used to transform raw cybersecurity network traffic data into meaningful and actionable security insights.

The final solution combines:

- Interactive dashboard storytelling
- Cyber threat intelligence analytics
- SOC-style monitoring experience
- Risk investigation capabilities
- Enterprise dashboard design

The dashboard successfully simulates a modern cybersecurity monitoring platform capable of helping organizations analyse threats, investigate suspicious activities, and support data-driven security decisions.